

SHOPEE BARCODE & VIDEO MANAGEMENT SYSTEM

Software Requirements Specification (SRS)

Version 1.0
May 2025

Document Title	SRS - Shopee Barcode & Video Management System		
Project	Shopee Barcode Video	Version	1.0
Prepared By	Mẫn Ngọc Đắc	Date	May 2025
Status	Final	Review Date	June 2025

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Shopee Barcode & Video Management System. This system is a mobile application designed to streamline the order fulfillment process at Shopee warehouses and partner stores by enabling staff to scan order barcodes/QR codes and automatically record video proof of package handling.

This document is intended for the development team, project stakeholders, quality assurance engineers, and business analysts involved in the project.

1.2 Project Scope

The Shopee Barcode & Video Management System is a cross-platform mobile application (iOS and Android) developed using React Native (Expo) with a Python FastAPI backend. The system enables:

- Warehouse staff and store employees to authenticate securely via email and password
- Staff to scan QR codes/barcodes on Shopee order packages using the device camera
- Automatic video recording of the packing/handling process, linked to the scanned order ID
- Video upload to the backend server for storage and future retrieval
- Viewing of historical scan and video records
- Administrators to manage staff accounts, stores, video records, and customer complaints
- Configurable user settings including dark mode and language preferences

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
BA	Business Analyst
API	Application Programming Interface
JWT	JSON Web Token — used for authentication
QR Code	Quick Response Code — machine-readable barcode used to encode order IDs
CRUD	Create, Read, Update, Delete — basic data operations
FR	Functional Requirement
NFR	Non-Functional Requirement
Admin	System Administrator with full access to all features
Employee	Warehouse/store staff with limited access

1.4 References

- React Native Documentation: <https://reactnative.dev>
- Expo SDK 51 Documentation: <https://docs.expo.dev>
- FastAPI Documentation: <https://fastapi.tiangolo.com>

-
- SQLAlchemy ORM Documentation: <https://docs.sqlalchemy.org>

2. Overall Description

2.1 Product Overview

The Shopee Barcode & Video Management System is a mobile-first application that bridges the gap between physical package handling and digital record-keeping. In a typical Shopee fulfillment workflow, workers pick, pack, and ship orders. This system provides a simple scan-and-record mechanism so that every order handling event is captured on video, creating an audit trail for dispute resolution, quality control, and process monitoring.

2.2 Product Features Summary

Feature ID	Feature Name	Description
F-01	User Authentication	Secure login/register with JWT tokens
F-02	Barcode/QR Scanning	Real-time camera scanning of Shopee order QR codes
F-03	Video Recording & Upload	Record packing video linked to scanned order, auto-upload
F-04	Video History	Staff can view their previously recorded videos
F-05	Complaint Management	Staff raise complaints; admins manage and resolve them
F-06	Admin Dashboard	Full management of staff, stores, videos, and complaints
F-07	User Settings	Dark mode, language preference, profile management
F-08	Auto Backend Discovery	App automatically finds backend server on the local network

2.3 User Classes and Characteristics

2.3.1 Employee (Warehouse Staff)

Warehouse or store employees are the primary users of the mobile application. They interact with the system on a daily basis during order fulfillment shifts. These users are expected to have basic smartphone proficiency but are not technically skilled. The interface must be simple and intuitive.

- Can log in and manage their own profile
- Can scan order QR codes using the device camera
- Can record and upload packing videos
- Can view their own video history
- Can submit complaints for problematic orders
- Cannot access admin features or other users' data

2.3.2 Administrator

Administrators have full system access and are responsible for managing operational data and resolving issues. They use the admin dashboard available within the same mobile application.

- Can perform all employee actions

- Can manage staff accounts (view, delete)
- Can manage store information (view, update, delete)
- Can manage all video records (view, delete)
- Can manage and resolve complaints

2.4 Operating Environment

- Mobile Platform: iOS 14+ and Android 10+ via Expo Go or standalone build
- Backend Server: Python 3.13+ with FastAPI running on local network (port 8000)
- Database: MySQL via XAMPP (local deployment)
- Network: App and backend must be on the same Wi-Fi network for auto-discovery
- Development Framework: React Native with Expo SDK 51

2.5 Assumptions and Dependencies

- The mobile device has a functional camera capable of video recording and QR scanning
- The backend server is running and accessible on the same local network
- MySQL database is properly initialized and running via XAMPP
- Users have a stable Wi-Fi connection during operation
- Expo Go app is installed on the device for development testing

3. Functional Requirements

3.1 Authentication Module

FR-01: User Registration

Requirement ID	FR-01
Feature	User Registration
Description	The system shall allow new users to register with name, email, password, and role (employee/admin). Duplicate email addresses must be rejected.
Input	Name (string), Email (valid email format), Password (string), Role (optional, default: employee)
Output	User object with user_id, name, email, role. Error 400 if email already registered.
Priority	High

FR-02: User Login

Requirement ID	FR-02
Feature	User Login
Description	The system shall authenticate users using email and password. Upon successful login, a JWT access token shall be returned along with user profile information.
Input	Email (string), Password (string)
Output	JWT access_token, token_type, user object (user_id, name, email, role). Error 400 for invalid credentials.
Priority	High

3.2 Barcode Scanning Module

FR-03: QR Code / Barcode Scanning

Requirement ID	FR-03
Feature	QR Code Scanning
Description	The system shall use the device's rear camera to continuously scan for QR codes. When a QR code is detected, the encoded order ID shall be displayed on screen and stored in state for the subsequent video recording.
Precondition	User is logged in; camera and microphone permissions are granted.
Output	Order ID string displayed; recording button becomes enabled.
Priority	High

3.3 Video Recording Module

FR-04: Video Recording

Requirement ID	FR-04
Feature	Video Recording
Description	After scanning a QR code, the user can start video recording. A timer shall display the elapsed recording duration. The user can stop recording at any time. Video is automatically saved as MP4 format.
Precondition	QR code has been scanned; camera is ready.
Output	Video file (MP4) saved locally, ready for upload.
Priority	High

FR-05: Video Upload

Requirement ID	FR-05
Feature	Video Upload
Description	Upon stopping the recording, the system shall automatically upload the video file to the backend server. The upload includes: video file (multipart), order_id, user_id, and start_time metadata.
Upload Timeout	60 seconds maximum
Output	Success: video_id, file_name, storage_url returned. Error alert on failure.
Priority	High

3.4 History Module

FR-06: View Scan & Video History

Requirement ID	FR-06
Feature	View History
Description	Logged-in employees can view a list of all their previously recorded videos, including order ID, file name, and recording timestamps.
Data Displayed	Order ID, Video file name, Start time, End time, Created date
Priority	Medium

3.5 Complaint Module

FR-07: Submit Complaint

Requirement ID	FR-07
Feature	Submit Complaint
Description	Authenticated employees can submit complaints for orders with issues. Issue types include: missing_item, wrong_item, other. A description is required.
Input	order_id, issue_type (enum), description (text)
Output	Complaint record created; complaint_id returned.
Priority	Medium

3.6 Admin Module

FR-08: Manage Staff Accounts

Requirement ID	FR-08
Feature	Manage Staff
Description	Admin can view all staff accounts and delete employee accounts from the system.
Priority	High

FR-09: Manage Stores

Requirement ID	FR-09
Feature	Manage Stores
Description	Admin can view all store records, update store details (name, email, phone, location), and delete stores.
Priority	Medium

FR-10: Manage Videos and Resolve Complaints

Requirement ID	FR-10
Feature	Manage Videos & Complaints
Description	Admin can view all uploaded video records, delete videos (removes file and metadata), view all complaints, mark complaints as resolved, and delete complaint records.
Priority	High

4. Non-Functional Requirements

4.1 Performance Requirements

ID	Requirement	Metric	Target
NFR-01	API Response Time	95th percentile	< 500ms for standard endpoints
NFR-02	Video Upload Time	Max timeout	60 seconds for videos up to 100MB
NFR-03	QR Scan Speed	Detection latency	< 1 second from code entering frame
NFR-04	Backend Discovery	Subnet scan time	< 10 seconds on typical network

4.2 Security Requirements

- NFR-05: All passwords must be hashed using bcrypt before storage; plain-text passwords must never be stored.
- NFR-06: All authenticated endpoints must validate JWT tokens before processing requests.
- NFR-07: File upload paths must be sanitized to prevent path traversal attacks (special characters replaced with underscores).
- NFR-08: Order IDs exceeding 255 characters must be SHA-256 hashed to prevent storage overflow.

4.3 Usability Requirements

- NFR-09: The mobile interface must be operable with one hand; primary actions (scan, record, stop) must be accessible from the main screen.
- NFR-10: The application must support dark mode to reduce eye strain during night-shift operations.
- NFR-11: Error messages must be displayed in clear, user-friendly language via custom alert dialogs, not raw system errors.
- NFR-12: The application must support English and Vietnamese language options.

4.4 Reliability Requirements

- NFR-13: The system must handle camera permission denial gracefully and display an appropriate message.
- NFR-14: If video upload fails, the user must receive a clear error notification. The application must not crash on upload failure.
- NFR-15: The backend discovery mechanism must fall back to a configurable URL if automatic scanning fails.

4.5 Scalability Requirements

- NFR-16: The database schema must support video file names up to 512 characters to accommodate long Shopee order IDs.

-
- NFR-17: The API must be stateless (JWT-based) to allow horizontal scaling of backend instances.

5. System Architecture Overview

5.1 Technology Stack

Layer	Technology	Purpose
Frontend	React Native + Expo SDK 51	Cross-platform mobile application
UI Styling	Tailwind CSS (NativeWind)	Consistent, responsive styling
HTTP Client	Axios	REST API calls, multipart file upload
Backend	Python 3.13 + FastAPI	REST API server, business logic
ORM	SQLAlchemy	Database abstraction layer
Database	MySQL (via XAMPP)	Persistent data storage
Authentication	JWT + bcrypt	Secure token-based auth, password hashing
File Storage	Local filesystem (/uploads/videos/)	Video file storage on backend server

5.2 API Endpoint Summary

Method	Endpoint	Description	Auth Required
POST	/register	Register new user	No
POST	/login	User login, returns JWT	No
PUT	/users/{user_id}	Update user profile	Yes
PUT	/users/{user_id}/password	Change password	Yes
GET	/users/{user_id}/store	Get user's store info	Yes
POST	/videos/upload	Upload video file + metadata	Yes
GET	/videos/me	Get current user's videos	Yes
GET	/videos	Get all videos (admin)	Yes
DELETE	/videos/{video_id}	Delete video + file	Yes
POST	/complaints	Submit a complaint	Yes
GET	/complaints	Get all complaints (admin)	Yes
PATCH	/complaints/{id}/resolve	Mark complaint resolved	Yes
GET	/staff	Get all employees (admin)	Yes
DELETE	/staff/{user_id}	Delete staff account (admin)	Yes

6. Data Requirements

6.1 Data Entities

- User: Stores user credentials, role (employee/admin), and store assignment
- Order: Stores Shopee order ID, customer details, and order status
- Product: Stores product details linked to an order including barcode
- Video: Stores video metadata (file name, storage URL, timestamps, linked order and user)
- Complaint: Stores complaint records linked to orders (issue type, description, resolution status)
- UserSettings: Stores per-user preferences (dark mode, language)
- StoreInfo: Stores store details (name, email, phone, location)

6.2 Data Retention

- Video files are stored indefinitely until manually deleted by an administrator
- Complaint records are retained until manually deleted by an administrator
- User settings persist across sessions using server-side storage

7. Constraints and Limitations

- The current implementation uses local filesystem storage for videos; cloud storage (e.g., AWS S3) is not implemented in v1.0
- The backend auto-discovery scans only the /24 subnet (254 hosts); larger networks are not supported
- The application requires both the mobile device and backend server to be on the same Wi-Fi network
- Admin role assignment is done at registration; there is no in-app role promotion workflow
- The system does not currently implement refresh tokens; expired JWT tokens require re-login

8. Appendix

8.1 Revision History

Version	Date	Author	Description
1.0	May 2025	Business Analyst	Initial SRS document created